Feedback-Augmented RT-DETR

for Small Object Detection

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Final Project — April 2026

Computer Vision and Image Processing
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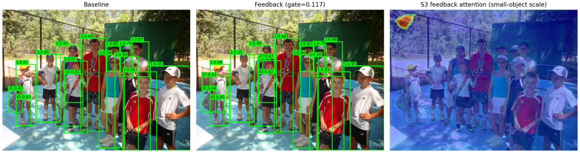
Problem Formulation

Small objects are harder to detect



AP_L (large)

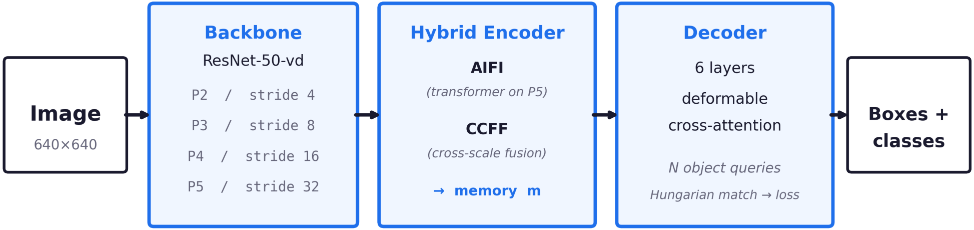
67.7

AP_S (small)

34.7

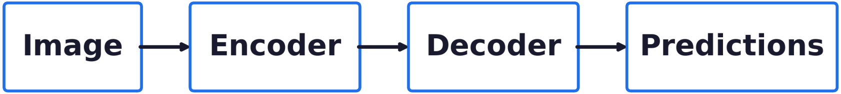
33-point gap. Why?

RT-DETR: backbone → encoder → decoder



End-to-end, NMS-free. Six decoder layers all attend over the same encoder memory.

The reason is structural



one-way information flow

Encoder memory is computed once. The decoder cannot revise it.

Refinement of the features small objects live in is a one-way street.

Data Sourcing Strategy

COCO 2017 — the standard small-object benchmark

Dataset

Microsoft COCO 2017

Splits

`train2017` — 118k images, 860k boxes

`val2017` — 5k images (held out)

Pre-trained weights

`rtdetr_r50vd_6x_coco.pth`

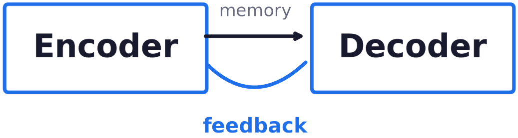
(public, finetuned 13 ep)

Why COCO?

- Standard benchmark — directly comparable to RT-DETR, DETR, YOLO.
- Has explicit small / medium / large size buckets (AP_S / AP_M / AP_L).
- $\sim$41% of annotated boxes are small (area $< 32^2$ px).
- Same pre-training corpus as our baseline $\rightarrow$ fair comparison.

Proposed Solution

Our idea: let the model refine its own features



Feed early decoder predictions back into the encoder memory.

Attention, in one line

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{Q K^\top}{\sqrt{d}}\right) V$$

Q	K	V
what we look for	where we look	what we extract
(query)	*(addresses)*	*(contents)*

Attention lets the model focus on important parts of the image.

Our feedback rule

$$\mathrm{m}' \;=\; \mathrm{m} \;+\; g \cdot \mathrm{Attn}(\mathrm{m}, \mathrm{t})$$

m

encoder memory

(image features)

g

gate

(how much feedback)

t

decoder output

(early predictions)

We update the encoder using the decoder's own predictions.

First attempt (v1) — and why it failed

Plain sigmoid gate, applied to all four pyramid levels.

$$\Delta \mathrm{AP}_S = 0.00$$

the mechanism contributed nothing

Why?

α drifts to $-\infty$ $\rightarrow$ gate ≈ 0.

The feedback signal is multiplied by zero before reaching memory.

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v2: two fixes — zero new parameters

1. Gate floor

$$g_{\text{eff}} = 0.1 + 0.9\,\sigma(\alpha)$$

feedback can never be silenced

2. P2 / P3 only

refine where small objects live

skip P4, P5 — focus the gradient

Reparameterize the gate. Refine only the levels small objects live in.

Performance Evaluation Approach

How we evaluate the contribution

Metrics (COCO standard)

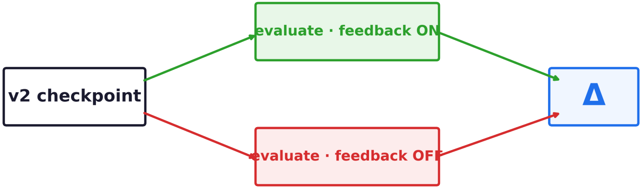
AP, AP_S, AP_M, AP_L — IoU 0.5..0.95

AP_S is the headline metric (area $< 32^2$ px)

Baselines

- RT-DETR-R50 (published)
- v1 feedback (before the fix)
- **v2 feedback** (with the fix)

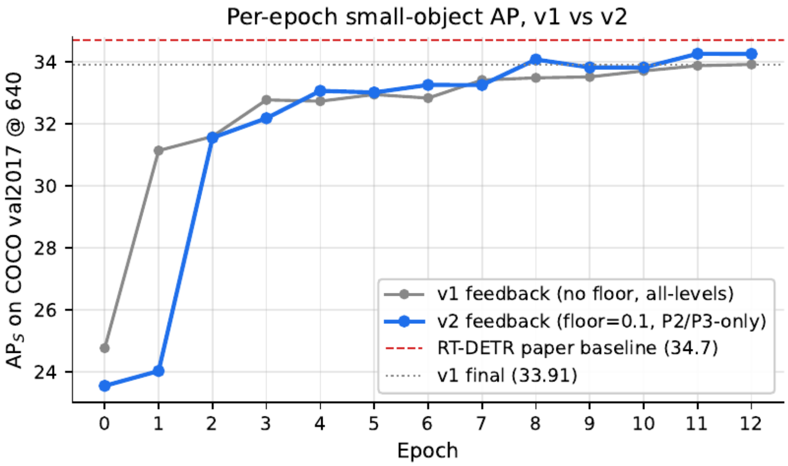
Same-checkpoint ablation — isolates causal contribution



Toggle feedback ON / OFF at inference. Identical weights, identical inputs.

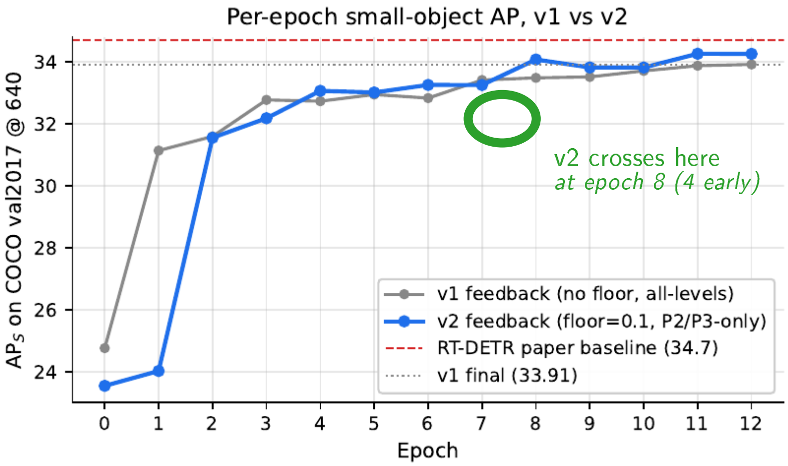
Final Results

Training: v2 climbs faster, ends higher



v2 first crosses v1's final 33.91 at epoch 8 — four epochs early.

Training: v2 climbs faster, ends higher



v2 first crosses v1's final 33.91 at epoch 8 — four epochs early.

Final performance (with feedback ON)

v1

33.91

AP_S

v2

34.30

AP_S +0.4

Causal effect of feedback

ON	OFF	Δ
34.25	33.26	+0.99
feedback enabled	*feedback bypassed*	*the contribution*

Feedback has a real causal impact on small-object detection.

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feedback enabled

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Δ
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Feedback has a real causal impact on small-object detection.

Higher resolution → bigger gain

640 × 640

34.25

→

800 × 800

37.01

+2.76 from more pixels.

Feedback helps. Spatial sampling helps too.

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Strengths and limitations

+ Strengths

- Causal contribution: $+\mathbf{0.99}$ on small-object AP, same checkpoint.
- Zero new parameters relative to v1.
- Consistent improvement on every metric (AP, S, M, L).
- Robust at higher resolution (37.01 small-object AP at 800×800).

– Limitations

- Single seed — multi-seed variance unmeasured.
- 13-epoch finetune (vs 72-epoch from scratch).
- Floor and mask not isolated (a 2×2 ablation is missing).
- $\approx 2\times$ slower than baseline ($53 \rightarrow 26$ FPS) — cost is the P2 level.

Future work: recovering inference speed

The 2× slowdown comes from the P2 level, not the feedback module.

1. Train with P2, infer without

fastest path

How. Use P2 during training, remove it at inference.

Goal. Keep most of the +0.99 gain without the P2 cost.

2. Lighter P2 path

compromise

How. Reduce P2's cost — fewer channels or simpler projection.

Goal. Preserve gains while reducing the high-resolution slowdown.

3. Knowledge distillation

principled

How. Distil v2's gains into a faster baseline RT-DETR student.

Goal. Recover baseline speed while keeping the AP_S gains.

Common goal: keep the +0.99 gain while approaching the 53 FPS baseline.

Conclusion

1. **Feedback improves small-object detection.**
   +0.99 small-object AP, causal contribution at inference.
2. **Only works with proper design.**
   Floor the gate; restrict to small-object levels.
3. **Modest but real.**
   Consistent across metrics; reproducible from the same checkpoint.

Thank you.

Questions?

Hatem Saadallah · Nour Jennane

github.com/HatemSaadallah/feedback-augmented-rtdetr